

SAMADHAN MANE

AI / Machine Learning Engineer | LLM & RAG Systems | Python
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SUMMARY

AI/ML engineer and final-year B.Tech Computer Engineering student specializing in agentic AI, RAG systems and deep learning. Winner of Neobim Hackathon 2026 (1st place) for ABYSS, a multi-tenant agentic RAG due-diligence platform. Builds end-to-end AI systems, from LangGraph tool-calling agents and hybrid retrieval to PyTorch audio models and speech-to-text pipelines, backed by full-stack backends with RBAC and concurrency-safe workflows.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, C++

Agentic AI & GenAI: Agentic AI, LangGraph, LangChain, LLM Tool Calling, RAG, Prompt Engineering, Reranking

ML, DL & NLP: PyTorch, Machine Learning, Deep Learning, NLP, Transformers, Hugging Face, Fine-tuning, Embeddings, TensorFlow, Keras, Scikit-learn, NumPy, Pandas

Backend: FastAPI, REST APIs, JWT, RBAC

Databases: Pinecone, ChromaDB, MongoDB, Redis

Cloud & DevOps: AWS, Docker, Git, GitHub

PROJECTS

ABYSS – Multi-Tenant AI Due Diligence Platform (Agentic RAG)

Mar 2026 – Apr 2026

Python, FastAPI, React, TypeScript, Pinecone, Redis, PostgreSQL, Gemini 3.5 Flash Lite, Docker | github.com/samadhanmane/Abbyss-ai | abyss-ai-gray.vercel.app

- Architected a **6-stage agentic RAG pipeline** (query routing, hybrid retrieval, generation, claim verification, page-level citation grounding) for multi-tenant document Q&A.
- Built an async ingestion pipeline (PyMuPDF, Tesseract OCR) for **15+ file types** with semantic chunking (1,200 chars, 150 overlap) and MiniLM embeddings stored in Pinecone.
- Combined dense search, BM25 and cross-encoder reranking (ms-marco-MiniLM) to reach **0.88 MRR** and **80.5% Recall@10** with retrieval + reranking in ~240 ms.
- Added guardrails (completeness gate with bounded retry, claim verification, citation grounding) for only ~180 ms overhead; **Redis caching** serves repeat queries in 12 ms vs ~3.7 s uncached.

BookMyHall – Multi-Tenant Campus Platform with Agentic AI Assistant

2025

LangGraph, LangChain, Gemini API (Tool Calling), React, Node.js, Express.js, MongoDB, JWT, Tailwind CSS | github.com/samadhanmane/BookMyHall | book-my-hall.vercel.app

- Built a **LangGraph agentic workflow** with Gemini tool calling that books halls and vehicles, places canteen orders and raises maintenance tickets from natural-language requests across 4 transactional workflows.
- Engineered concurrency-safe booking with 5-minute slot locks and race-condition prevention; **load-tested 1,005 requests** (28.3 req/s, P95 3.0 s) with 100% success on all non-booking scenarios and all 173 conflicting bookings correctly rejected with HTTP 409.
- Designed **10-tier RBAC** (Super Admin to Technician) with per-organization data isolation and multi-level approval chains (Faculty, HOD, Registrar, Director).
- Developed the full-stack platform (React user/admin portals, Express REST API, MongoDB) with real-time status tracking and asynchronous email notifications.

MOM-ai – Deep Learning Meeting Intelligence System (ASR + VAE + NLP + RAG)

2026

Python, PyTorch, Whisper, FLAN-T5, Mistral, LangChain, ChromaDB, Streamlit | github.com/samadhanmane/MOM-ai | meeting-assistant-mom.streamlit.app

- Engineered an **acoustic representation learning pipeline** in PyTorch, training Conv2D Autoencoder and VAE models on 64-band log-Mel spectrograms (16 kHz, 128 frames) to achieve **25.34 dB PSNR**, **0.974 SSIM** and 0.00293 MSE on 256-dim latents with 14 ms inference latency.
- Architected a multi-task extraction pipeline using **OpenAI Whisper ASR** and **Google FLAN-T5** to transcribe and synthesize meetings into structured summaries, decisions and action items, scoring **0.8641 BERTScore** and 48.6% ROUGE-L on Edinburgh AMI Corpus slices.
- Designed a multi-model routing & **RAG system** in LangChain LCEL with ChromaDB and Mistral-Embed, routing audio between Whisper and Sarvam AI for Hinglish code-mixed speech and cutting context token overhead by ~65% with sub-1.5 s retrieval latency.

OPEN-SOURCE CONTRIBUTION

PipesHub AI – Reasoning-Effort Defaults for LLM Models (Issue #3201)

2026

Python, LLM Gateway, Unit Testing | [PR #3290](https://github.com/samadhanmane/PipesHub/pull/3290) | [Follow-up PR #3828 \(merged\)](https://github.com/samadhanmane/PipesHub/pull/3828)

- Diagnosed reasoning models failing whenever no effort was passed because the platform hardcoded a **"high"** fallback that some models reject; authored PR #3290 with a 3-tier resolution (**request-level override, then model-level default, then platform fallback**) and unit tests for nested configs, request overrides and non-reasoning models.
- Maintainers built on this approach in merged PR #3828, extending it to schema persistence, validation, Bedrock support, health checks and frontend payloads, with attribution to #3290.

EDUCATION

B.Tech, Computer Engineering – MIT Academy of Engineering, Pune | CGPA: 7.49/10

2023 – 2027 (Exp.)

ACHIEVEMENTS

- Winner (1st Place), Neobim Hackathon 2026** – INR 1.8 lakh prize pool; 4-member team; ABYSS recognized for its agentic architecture and production readiness.